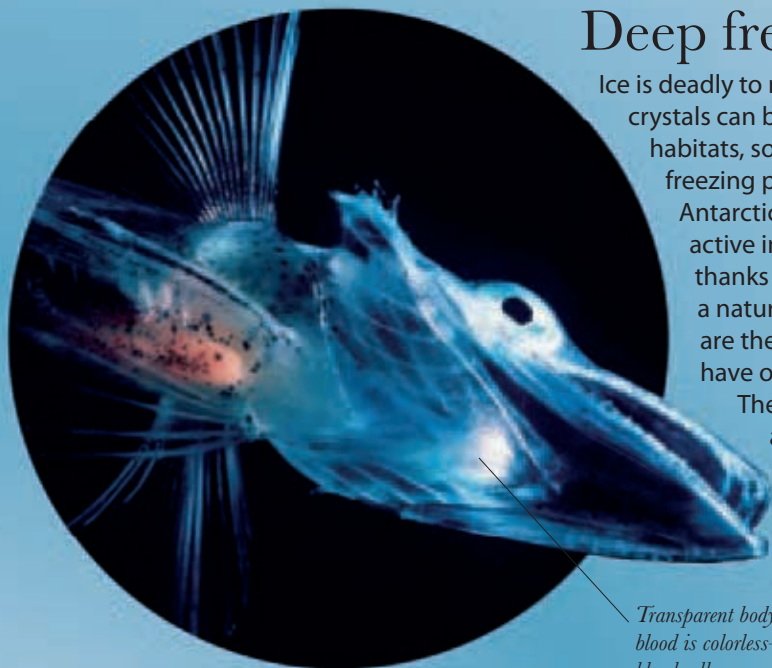


Deep freeze

Ice is deadly to most animals, because its crystals can burst living cells. But in cold habitats, some animals survive below freezing point. Among them are Antarctic icefish, which can stay active in seawater at 28°F (−2°C), thanks to proteins that work like a natural antifreeze. Some icefish are the only vertebrates that do not have oxygen-carrying red blood cells. The fluid in their blood carries all the oxygen they need.



Transparent body shows that icefish blood is colorless—it contains no red blood cells



Droplet of condensation on bat's cold body

Leg rest

Microscopic animals can be amazingly good at coping with extreme conditions. Tardigrades stow their legs away if conditions turn dry. Once they have done this, they show no signs of life. They can remain sealed up for 10 years or more.



Active tardigrades with legs extended

Waxy covering produced by tardigrade prevents water loss when dormant



Dormant tardigrade with legs retracted

Chilling out

Deep in a cave, a hibernating whiskered bat waits for winter to come to an end. Its body temperature is just a few degrees above freezing, and droplets of water have condensed on its fur. During winter, the bat wakes up for short periods and shivers, which helps to keep its muscles functioning normally.

Skin cocoon peels away as frog emerges



Waiting for rain

In Australia, the water-holding frog survives in deserts by holing up underground. It stores water in its bladder and seals itself in a cocoon of dead, moisture-retaining skin. After heavy storms, it digs its way to the surface, where it lays its eggs in temporary pools. The pools soon dry, but by then, the frogs and froglets are safely hidden away.

Hazelnut stored by dormouse for eating when it wakes briefly from hibernation

